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In [1]: # Program 7: Correlation matrix and heatmap
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

FILE = r"D:\Datascience-MachineLearning\Datasets\iris_dataset.csv"    # <-- change to your file path

df = pd.read_csv(FILE)
corr = df.select_dtypes(include="number").corr()

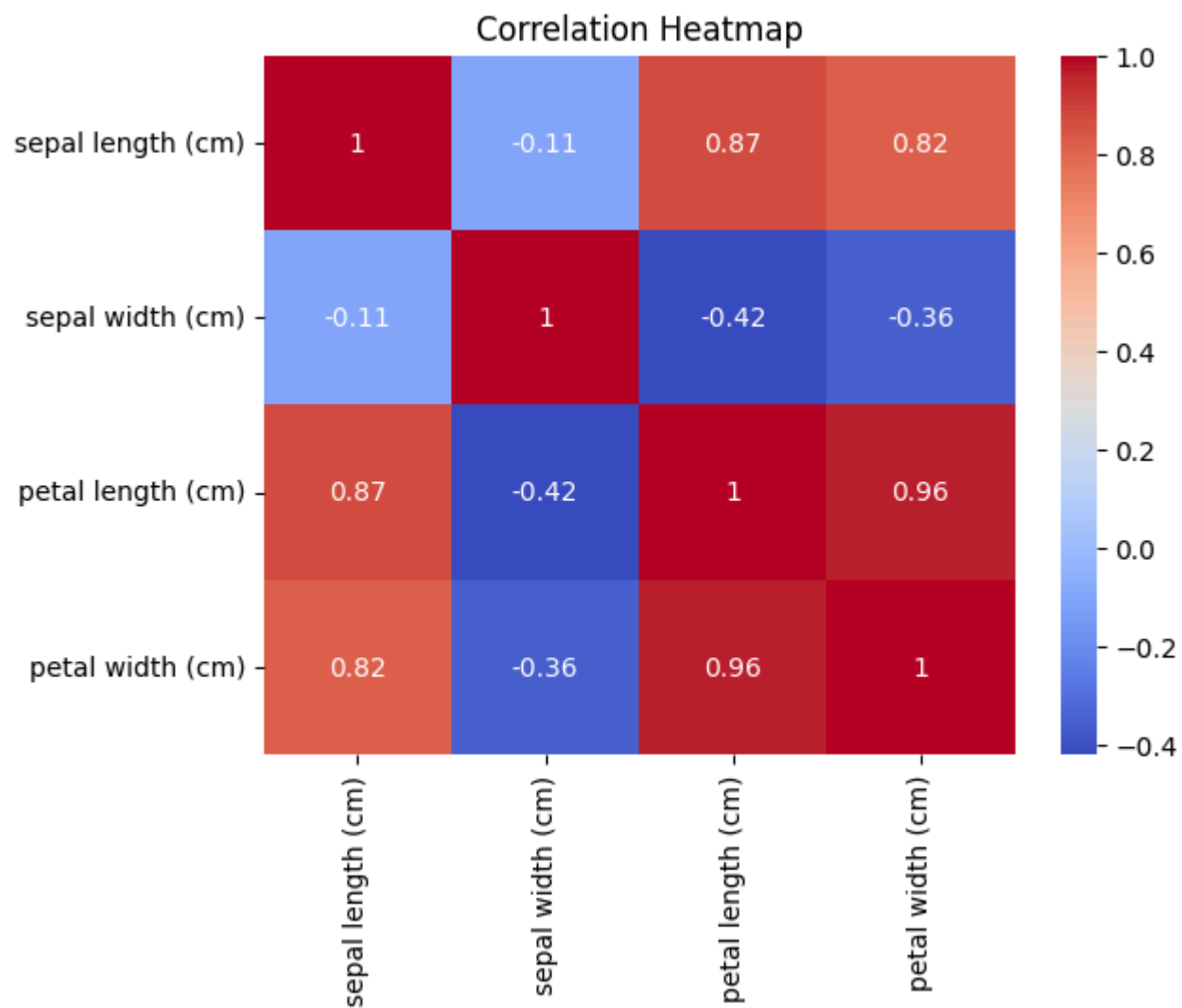
print(corr)

sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()

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	sepal length (cm)	sepal width (cm)	petal length (cm)	\
sepal length (cm)	1.000000	-0.109369	0.871754	
sepal width (cm)	-0.109369	1.000000	-0.420516	
petal length (cm)	0.871754	-0.420516	1.000000	
petal width (cm)	0.817954	-0.356544	0.962757	

	petal width (cm)
sepal length (cm)	0.817954
sepal width (cm)	-0.356544
petal length (cm)	0.962757
petal width (cm)	1.000000



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In [2]: # Program 7: Correlation matrix and heatmap
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

FILE = r"D:\Datascience-MachineLearning\Datasets\diamonds.csv"    # <-- change to your file path
```

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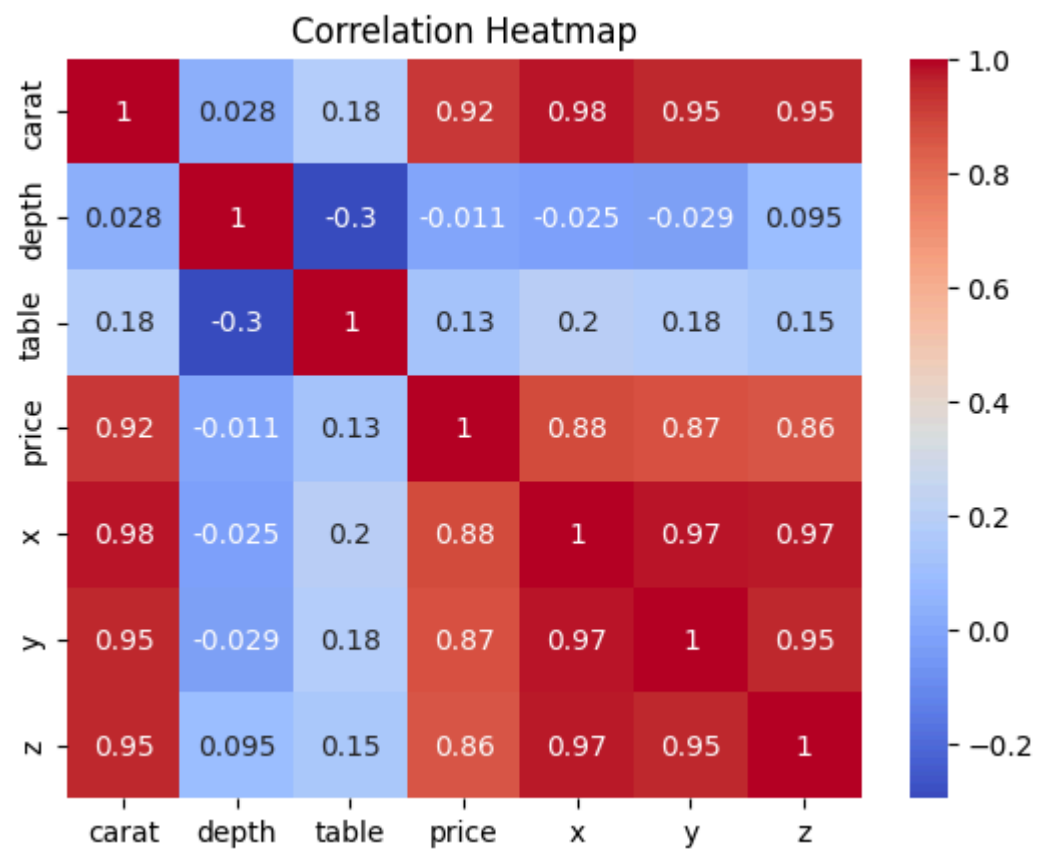
df = pd.read_csv(FILE)
corr = df.select_dtypes(include="number").corr()

print(corr)

sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()

```

	carat	depth	table	price	x	y	z
carat	1.000000	0.028224	0.181618	0.921591	0.975094	0.951722	0.953387
depth	0.028224	1.000000	-0.295779	-0.010647	-0.025289	-0.029341	0.094924
table	0.181618	-0.295779	1.000000	0.127134	0.195344	0.183760	0.150929
price	0.921591	-0.010647	0.127134	1.000000	0.884435	0.865421	0.861249
x	0.975094	-0.025289	0.195344	0.884435	1.000000	0.974701	0.970772
y	0.951722	-0.029341	0.183760	0.865421	0.974701	1.000000	0.952006
z	0.953387	0.094924	0.150929	0.861249	0.970772	0.952006	1.000000



In []: